

Trainings ▾

General Information

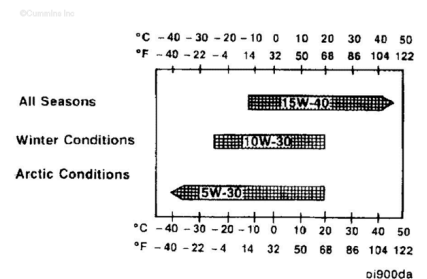
Use Cummins Premium Blue® 15W-40 multiviscosity oil, or equivalent, that meets API Classification CE for turbocharged engines.

Note : CC/CD or CD/SF engine oils can be used in areas where CE oil is **not** yet available; however, the oil drain interval must be reduced by 50 percent.



⚠ CAUTION ⚠

Limited use of low-viscosity oils, such as 10W-30, can aid in starting the engine and providing sufficient oil flow at ambient temperatures below -5°C [23°F]. However, continuous use of low-viscosity lubricating oils can decrease engine life. Refer to the accompanying chart.

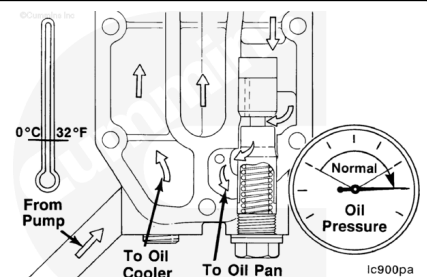


Lubricating Oil System - Diagnosing Malfunctions

Make sure to check items related to oil pressure, such as gauges, high and low oil level, excessive contamination, and oil viscosity.

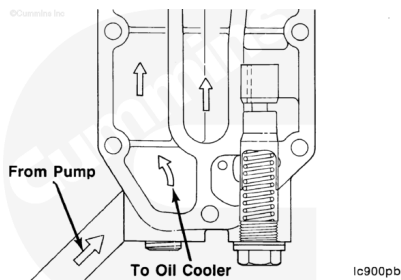
High Oil Pressure

High oil pressure usually occurs after the engine is first started in cold weather. The lubrication system does **not** have a cold start relief valve. The pressure regulating valve components are machined to a size that will relieve the excessive pressure created by cold engine oil.



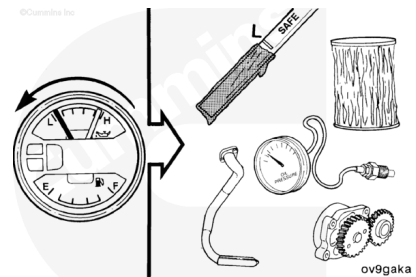
Oil Pressure Regulating Valve

The engine will have high oil pressure if the regulator sticks shut. Inspect the regulator for freedom of movement.



Low Oil Pressure

Low oil pressure can be caused by several lubrication system-related malfunctions. To begin the investigation, determine the engine operating conditions when the low pressure was first observed.

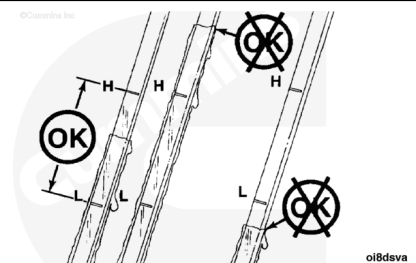


Oil Level

An improper lubricating oil level can cause low lubricating oil pressure.

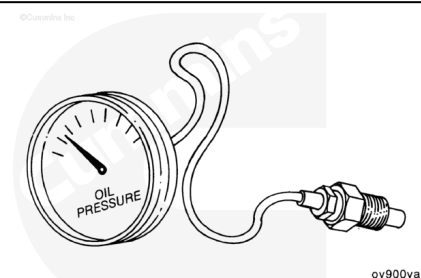
⚠ CAUTION ⚠

Never operate the engine with the oil level below the low (L) mark or above the high (H) mark.



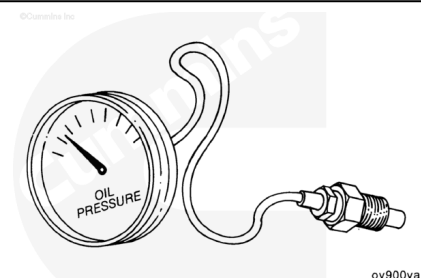
Oil Filter

A plugged filter will cause a gradual loss of oil pressure by approximately 69 kPa [10 psi]. This will cause the bypass valve to open, allowing unfiltered oil to flow to internal engine components. The oil pressure will remain low until a new filter is installed.



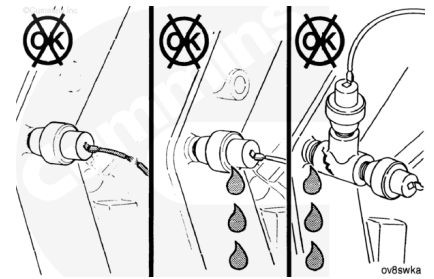
Oil Pressure Gauge - Inspection

Inspect the oil gauge and sending unit to make sure they are operating correctly by verifying the pressure with manual gauge.



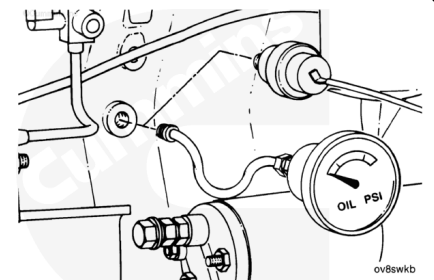
Inspect for the following defects:

- Electrical wiring broken
- Sending unit malfunction
- Plumbing loose or broken.



If a sending unit malfunction is found:

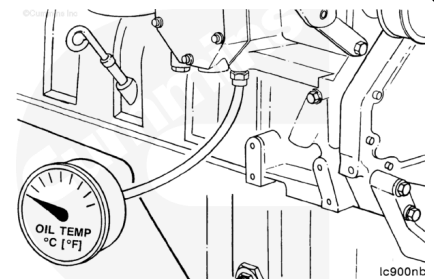
- Use a master gauge of known accuracy to verify the reading of the suspect gauge.
- Connect the line from the master gauge to the main oil rifle on the fuel pump side of the engine.
- Replace the sending unit if it is defective.



Oil Temperature Gauge - Inspection

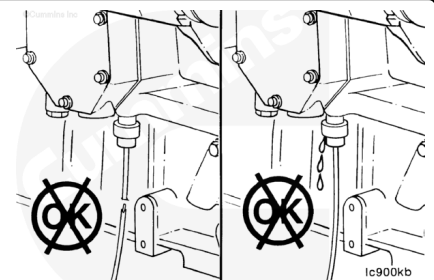
Remove the pipe plug from the bottom of the oil cooler, and install the oil temperature gauge sending unit.

celsius		fahrenheit
126.6	MAX	260



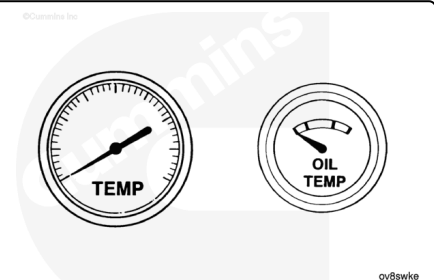
Inspect for the following defects:

- Electrical wiring broken
- Sending unit malfunction.



If a sending unit malfunction is found:

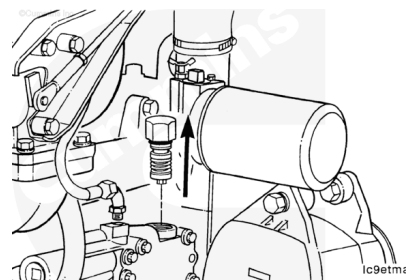
- Use a master gauge of known accuracy to verify the reading of the suspect gauge.
- Replace the sending unit if it is defective.



Oil Temperature Thermostat

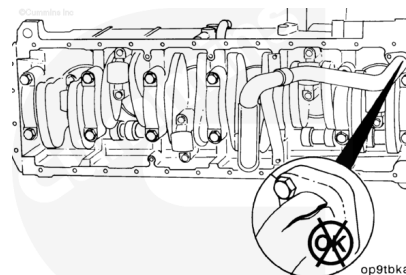
Note : The oil temperature thermostat can not be checked in-chassis; it must be removed for testing;

refer to Procedure 007-039
(/qs3/pubsys2/xml/en/procedures/41/41-007-039.html).



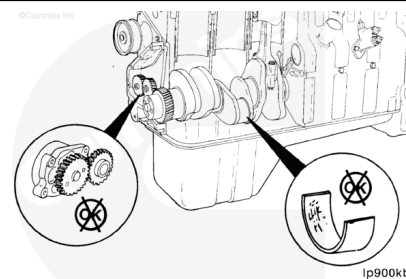
Oil Suction Tube

A loose suction tube, damaged gasket, or crack in the suction tube can cause a temporary loss of prime for the oil pump. The engine will have low pressure or no oil pressure at starting, followed by normal or low pressure.



Bearings and Oil Pump

A steady decrease in oil pressure over a long period will be indicated by worn bearings or excessive oil pump wear. Refer to Procedure 007-051
(/qs3/pubsys2/xml/en/procedures/41/41-007-051.html) to check for internal engine damage.

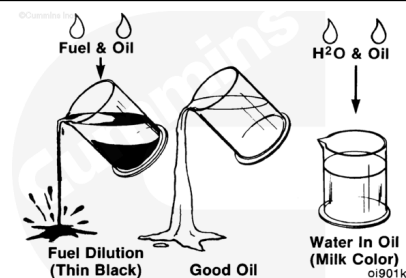


Oil Dilution

Diluted oil can cause severe engine damage.

Inspect the condition of the oil.

- Thin, black oil indicates fuel in the oil.
- Milky discoloration is an indication of coolant in the oil.



Coolant-Diluted Oil

Coolant in the oil results from a crack or leak between the coolant and oil circuits.

The possibility for intermixing can occur in these components:

- Oil cooler
- Aftercooler
- Core plugs in cylinder head
- Cylinder liner seals
- Head gasket
- Cylinder liner (crack)

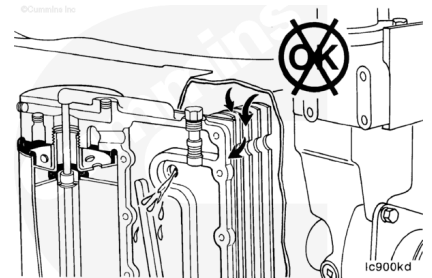
- Cylinder head (cracked passage)
- Cylinder block (cracked passage)
- Air compressor (coolant cooled).

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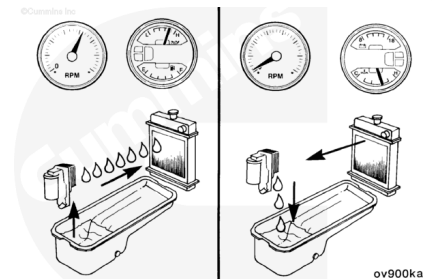
Oil Cooler

The oil cooler design does **not** require gaskets or seals to maintain the separation of oil and coolant.

If either the coolant or oil is contaminated, check for a leaking oil cooler element; refer to Procedure 007-007 (</qs3/pubsys2/xml/en/procedures/41/41-007-007.html>).



During operation the oil pressure will be higher than coolant pressure. A leak in the oil cooler will show as oil in the coolant. However, following an engine shutdown, the residual pressure in the cooling system can cause coolant to seep through the leak path into the oil.

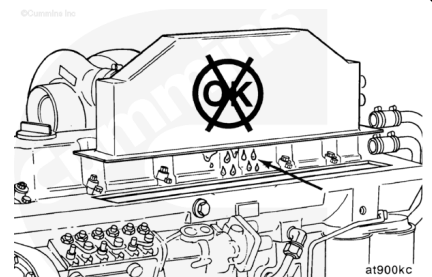


Aftercooler

The aftercooler is also a source from which coolant can leak into the lubricating oil.

Remove the aftercooler and look for evidence of leaking into the intake manifold.

Pressure test the aftercooler element; refer to Procedure 010-008 (</qs3/pubsys2/xml/en/procedures/41/41-010-008.html>).

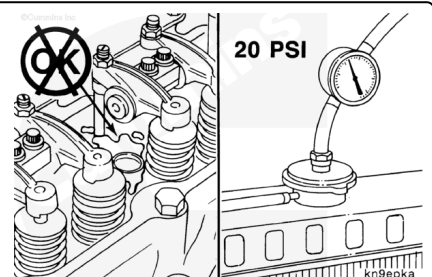


Cylinder Head Expansion Plugs

The expansion plugs in the cylinder head under the valve cover is another potential for oil dilution.

If possible, inspect for the leaks while the engine is warm. Remove the valve cover to look for signs of leaks.

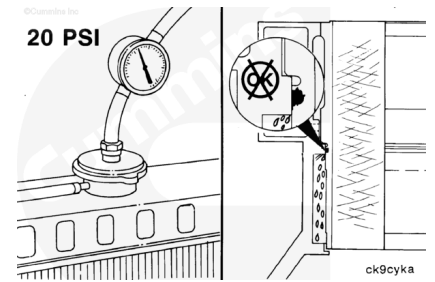
Pressurize the coolant system to 140 kPa [20 psi], if necessary.



Cylinder Liner Seals

Coolant can enter the lubricating oil through a deteriorated or damaged liner seal.

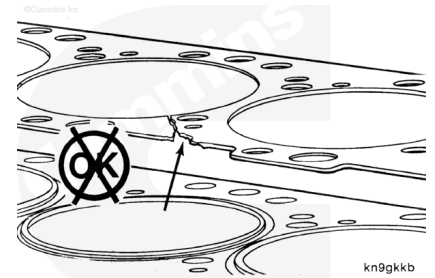
Remove the oil pan and inspect the bottom side of liners with the cooling system pressurized.



Cylinder Head Gasket

Coolant in the oil can also be caused by a damaged cylinder head gasket.

Pressurize the cooling system to check for leaks. Remove the oil pan to locate internal leaks, if necessary; refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/41/41-007-025.html>).



Cracked Cylinder Head

A crack in the head from the water jacket to an oil passage or to the top rocker lever area will cause oil dilution.

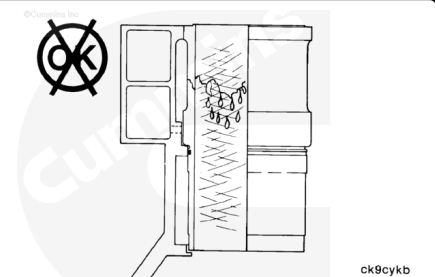
Pressurize the cooling system to 140 kPa [20 psi] and check for leaks.



Cracked Cylinder Liner

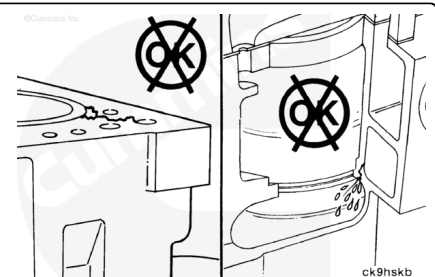
A cracked cylinder liner can leak coolant into the lubricating oil. Remove oil pan and look for coolant leaking from inside of liner bore.

Note : Air compressor leaks will produce the same symptoms; refer to Procedure 012-019 (</qs3/pubsys2/xml/en/procedures/41/41-012-019.html>) before concluding that the leak is from the cylinder liner.



Cracked Cylinder Block

A crack in the cylinder block from an oil drilling or passage to the water jacket can cause oil dilution and can normally be found either as an external leak from a gasket (i.e., head gasket), or from the oil pickup tube with the oil pan off.

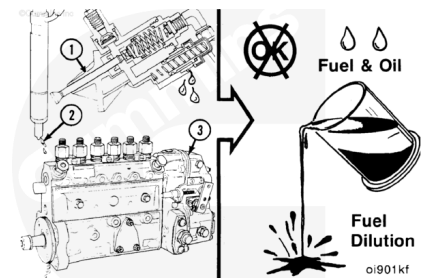


The cooling system **must** be pressurized to 140 kPa [20 psi] to detect leaks.

Fuel Diluted Oil

Fuel dilution can **only** come from three sources:

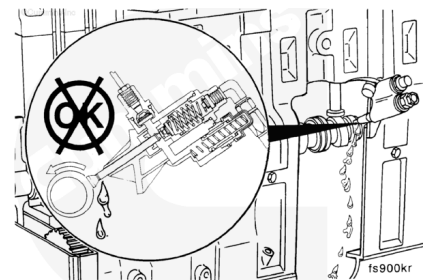
1. Fuel transfer pump
2. Fuel leaking over the rings
3. Injection pump internal wear.



Fuel Transfer Pump

Note : On nonautomotive engines, a worn or damaged seal around the plunger can allow fuel to leak into the oil pan.

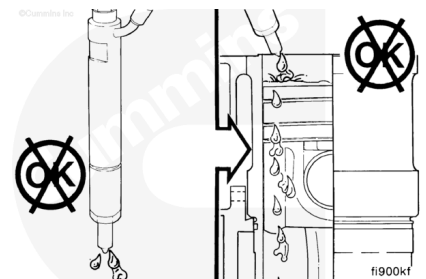
Automotive engines have a weep hole to allow the fuel to leak externally.



Fuel Leaking by Piston Rings

Incomplete combustion in the cylinders can result in unburned fuel draining into the oil pan.

This condition can be caused by a leaking injector or reduced compression caused by inadequate piston ring sealing.

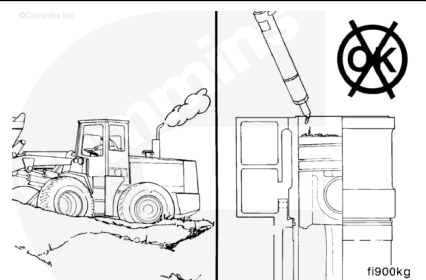


An increase in white exhaust smoke during the first start of the day is a symptom of an injector leaking.

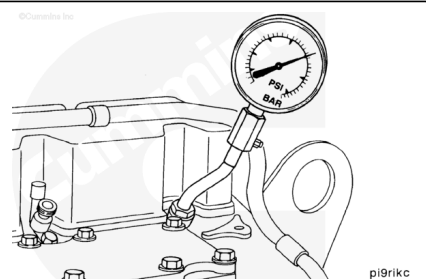
An injector leak will also cause the engine to run rough and have low power.

Remove and replace leaking injectors; refer to Procedure 006-026 (</qs3/pubsys2/xml/en/procedures/41/41-006-026.html>).

Refer to the Shop Manual, C Series Engines, Bulletin No. 3810476, for test and repair instructions.

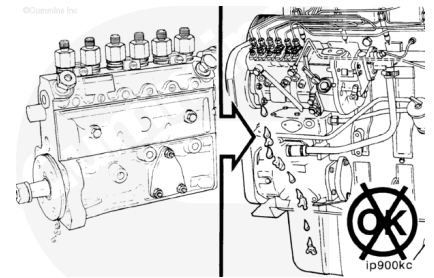


Perform a compression check to verify piston rings are properly sealed; refer to Procedure 001-999 (</qs3/pubsys2/xml/en/procedures/41/41-001-999.html>).



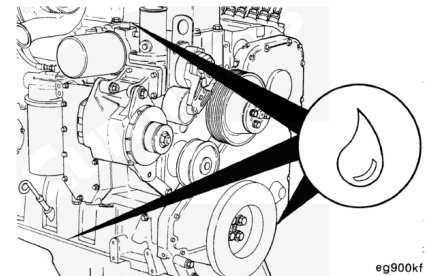
Injection Pump

A worn or damaged injection pump can allow fuel to leak into the lubricating oil as it passes through the pump.

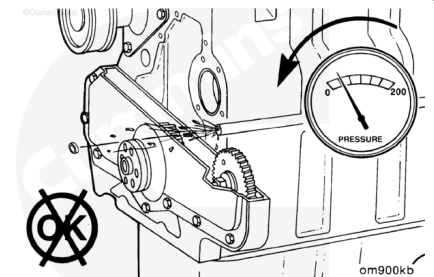


Oil Leaks

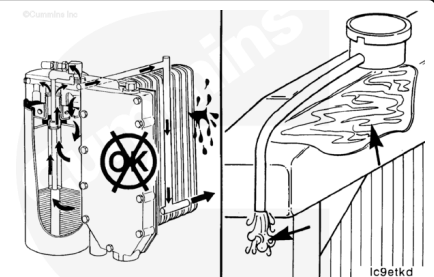
Various gaskets, seals, and plugs are used to contain the oil. Most leaks can be identified during routine inspection of the engine and vehicle.



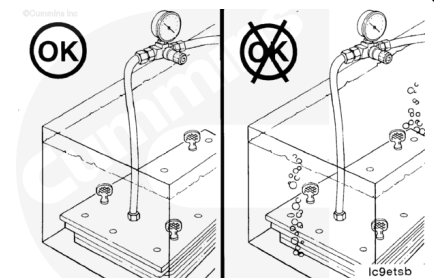
A damaged rocker assembly oil manifold or blown expansion plug can allow a large quantity of oil to escape, resulting in a sudden drop in the oil pressure.



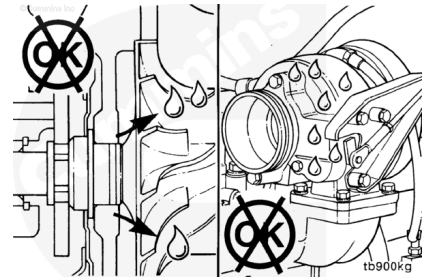
If the oil cooler element ruptures, the oil pressure will force oil into the cooling system. Oil in the coolant can be visible when the radiator cap is removed. As the oil is forced into the cooling system, coolant will be displaced through the radiator overflow.



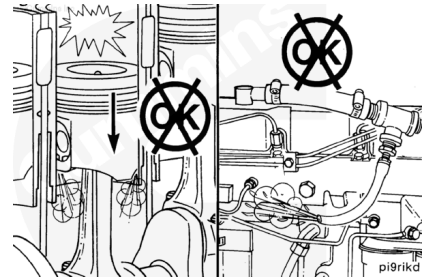
Leaks can be verified by pressure testing the oil cooler element with lubricating oil cooler pressure test kit, Part No. 3823876. Apply 483 kPa [70 psi] air pressure to the element to check for leaks.



High intake air restriction and worn or damaged seals in the turbocharger can allow oil to leak into the air crossover pipe and be burned in the engine. This condition can be verified by removing the air crossover tube and looking for oil.



Inadequate sealing of the piston rings will result in combustion gas and oil droplets being blown out the crankcase breather tube and/or consumed by the engine. Refer to Procedure 014-010 (</qs3/pubsys2/xml/en/procedures/41/41-014-010.html>) for measuring crankcase gases (blowby).



Oil can be lost through a worn or malfunctioning air compressor. Look for carbon buildup in the air line from the compressor to the air tank, and look for oil when draining the tank. Refer to Compressed Air for additional diagnostic procedures.

